

1. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 . You should show your working and/or justification clearly in all answers that involve calculations or derivations.

The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = \frac{1}{4}X_{t-1} - \frac{1}{48}X_{t-2} + Z_t$$

- (a) What type of process is $\{X_t : t \in \mathbb{Z}\}$ including its order?

SOLUTION

$\{X_t : t \in \mathbb{Z}\}$ is an autoregressive process of order 2.

- (b) Is the process $\{X_t : t \in \mathbb{Z}\}$ (weakly) stationary? Justify your answer.

SOLUTION

The process $\{X_t : t \in \mathbb{Z}\}$ can be written as

$$\begin{aligned} X_t - \frac{1}{4}X_{t-1} + \frac{1}{48}X_{t-2} &= Z_t \\ \implies \left(1 - \frac{B}{4} + \frac{B^2}{48}\right)X_t &= Z_t \\ \implies \phi(B)X_t &= Z_t \end{aligned}$$

where B is the backward shift operator and $\phi(z) = 1 - \frac{z}{4} + \frac{z^2}{48}$.

The process is (weakly) stationary if the roots of $\phi(z) = 0$ lie outside the unit circle.

We set

$$\begin{aligned} 1 - \frac{z}{4} + \frac{z^2}{48} &= 0 \\ \implies z &= \frac{\frac{1}{4} \pm \sqrt{\frac{1}{16} - \frac{4}{48}}}{\frac{1}{24}} \\ &= \frac{\frac{1}{4} \pm \frac{i}{\sqrt{48}}}{\frac{1}{24}} \\ &= 6 \pm i2\sqrt{3}. \end{aligned}$$

Then $|z| = \sqrt{48} > 1$ and, hence, the process is (weakly) stationary.

- (c) Write down the Yule-Walker equations for the process $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION

The Yule-Walker equations are

$$\rho(h) - \frac{1}{4}\rho(h-1) + \frac{1}{48}\rho(h-2) = 0.$$

- (d) By solving the Yule-Walker equations from (c), determine the autocorrelation function (ACF) of $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION

The auxiliary equation is

$$\lambda^2 - \frac{1}{4}\lambda + \frac{1}{48} = 0.$$

Solving the auxiliary equation, we obtain

$$\begin{aligned} \left(\lambda - \frac{1}{8}\right)^2 - \frac{1}{64} + \frac{1}{48} &= 0 \\ \implies \left(\lambda - \frac{1}{8}\right)^2 &= -\frac{1}{192} \\ \implies \lambda - \frac{1}{8} &= \pm \frac{i}{8\sqrt{3}} \\ \implies \lambda &= \frac{1}{8} \left(1 \pm i \frac{\sqrt{3}}{3}\right) \\ &= \frac{\sqrt{3}}{12} e^{\pm i \frac{\pi}{6}}. \end{aligned}$$

The auxiliary equation has complex roots and, therefore, the general solution has the form

$$\rho(h) = A \left(\frac{\sqrt{3}}{12}\right)^{|h|} e^{\frac{i\pi|h|}{6}} + C \left(\frac{\sqrt{3}}{12}\right)^{|h|} e^{\frac{-i\pi|h|}{6}}$$

where A and C are constants.

Since $\rho(0) = 1$ we have

$$A + C = 1.$$

Also

$$\begin{aligned} \rho(1) - \frac{1}{4}\rho(0) + \frac{1}{48}\rho(-1) &= 0 \\ \implies \frac{49}{48}\rho(1) &= \frac{1}{4} \\ \implies \rho(1) &= \frac{12}{49}. \end{aligned}$$

Then, for $h = 1$:

$$A \left(\frac{\sqrt{3}}{12}\right) e^{\frac{i\pi}{6}} + C \left(\frac{\sqrt{3}}{12}\right) e^{\frac{-i\pi}{6}} = \frac{12}{49}$$

$$\begin{aligned}
 \Rightarrow A \left(\frac{\sqrt{3}}{12} \right) e^{\frac{i\pi}{6}} + (1-A) \left(\frac{\sqrt{3}}{12} \right) e^{\frac{-i\pi}{6}} &= \frac{12}{49} \\
 \Rightarrow A \left[i \frac{\sqrt{3}}{12} \left(2 \sin \left(\frac{\pi}{6} \right) \right) \right] &= \frac{12}{49} - \left(\frac{\sqrt{3}}{12} \right) e^{\frac{-i\pi}{6}} \\
 \Rightarrow A \left(i \frac{\sqrt{3}}{12} \right) &= \frac{12}{49} - \frac{\sqrt{3}}{12} \left(\frac{\sqrt{3}}{2} - \frac{i}{2} \right) \\
 \Rightarrow A &= \frac{12}{i\sqrt{3}} \left[\frac{12}{49} - \frac{\sqrt{3}}{12} \left(\frac{\sqrt{3}}{2} - \frac{i}{2} \right) \right] \\
 &= \frac{1}{2} - i4\sqrt{3} \left(\frac{12}{49} - \frac{3}{24} \right) \\
 &= \frac{1}{2} - i \frac{47\sqrt{3}}{98}.
 \end{aligned}$$

Hence

$$C = \frac{1}{2} + i \frac{47\sqrt{3}}{98}.$$

and

$$\rho(h) = \left(\frac{1}{2} - i \frac{47\sqrt{3}}{98} \right) \left(\frac{\sqrt{3}}{12} \right)^{|h|} e^{\frac{i\pi|h|}{6}} + \left(\frac{1}{2} + i \frac{47\sqrt{3}}{98} \right) \left(\frac{\sqrt{3}}{12} \right)^{|h|} e^{\frac{-i\pi|h|}{6}}$$

which simplifies to

$$\begin{aligned}
 \rho(h) &= \left(\frac{1}{2} - i \frac{47\sqrt{3}}{98} \right) \left(\frac{\sqrt{3}}{12} \right)^{|h|} \left(\cos \left(\frac{\pi|h|}{6} \right) + i \sin \left(\frac{\pi|h|}{6} \right) \right) \\
 &\quad + \left(\frac{1}{2} + i \frac{47\sqrt{3}}{98} \right) \left(\frac{\sqrt{3}}{12} \right)^{|h|} \left(\cos \left(\frac{\pi|h|}{6} \right) - i \sin \left(\frac{\pi|h|}{6} \right) \right) \\
 &= \left(\frac{\sqrt{3}}{12} \right)^{|h|} \left[\cos \left(\frac{\pi|h|}{6} \right) + \frac{47\sqrt{3}}{49} \sin \left(\frac{\pi|h|}{6} \right) \right].
 \end{aligned}$$

(e) Hence determine the partial autocorrelation function (PACF) of $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION

We know that the PACF at lag 1 is equal to the ACF at lag 1. Thus

$$\rho(0, 1) = \frac{12}{49}.$$

Considering the process written in autoregressive form as

$$X_t = \frac{1}{4}X_{t-1} - \frac{1}{48}X_{t-2} + Z_t$$

the PACF at lag 2 is the coefficient of X_{t-2} (i.e. $-1/48$).

At lags > 2 , the PACF will be zero. This is because the PACF at lag m is the coefficient of X_{t-m} in the autoregressive model

$$X_t = \beta_{m,1}X_{t-1} + \beta_{m,2}X_{t-2} + \dots + \beta_{m,m}X_{t-m} + Z_t.$$

Since we have $X_t = \frac{1}{4}X_{t-1} - \frac{1}{48}X_{t-2} + Z_t$ it follows that $\beta_{m,m}$ will be zero for all $m > 2$.

Then, the PACF is

$$\gamma(0, m | \dots, m-1) = \begin{cases} \frac{12}{49} & m = 1; \\ -\frac{1}{48} & m = 2; \\ 0 & m > 2. \end{cases}$$

2. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 . You should show your working and/or justification clearly in all answers that involve calculations or derivations.

The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = Z_t + \frac{2}{5}Z_{t-1} + \frac{1}{4}Z_{t-2}.$$

- (a) Determine the autocovariance generating function of $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION

To determine the autocovariance generating function we write the process as

$$\begin{aligned} X_t &= \left(1 + \frac{2B}{5} + \frac{B^2}{4}\right) Z_t \\ &= \alpha(B)Z_t \end{aligned}$$

where B is the backward shift operator and $\alpha(B) = 1 + \frac{2B}{5} + \frac{B^2}{4}$.

Then the autocovariance generating function is

$$\begin{aligned} \Gamma(z) &= \sigma_z^2 \alpha(z) \alpha(z^{-1}) \\ &= \sigma_z^2 \left(1 + \frac{2z}{5} + \frac{z^2}{4}\right) \left(1 + \frac{2z^{-1}}{5} + \frac{z^{-2}}{4}\right) \\ &= \sigma_z^2 \left[\left(1 + \frac{4}{25} + \frac{1}{16}\right) + \left(\frac{2}{5} + \frac{2}{20}\right)z + \left(\frac{2}{5} + \frac{2}{20}\right)z^{-1} + \frac{z^2}{4} + \frac{z^{-2}}{4} \right] \\ &= \sigma_z^2 \left(\frac{489}{400} + \frac{1}{2}z + \frac{1}{2z} + \frac{z^2}{4} + \frac{1}{4z^2} \right) \end{aligned}$$

- (b) Derive the spectral density function of $\{X_t : t \in \mathbb{Z}\}$, expressing your answer in terms of $\cos(w)$ where $w \in (-\pi, \pi]$ is the frequency.

SOLUTION

Using the autocovariance generating function from (a), the spectral density function is

$$\begin{aligned} g(w) &= \frac{\sigma_z^2}{2\pi} \Gamma(e^{iw}) \\ &= \frac{\sigma_z^2}{2\pi} \left(\frac{489}{400} + \frac{1}{2} (e^{iw} + e^{-iw}) + \frac{1}{4} (e^{i2w} + e^{-i2w}) \right) \\ &= \frac{\sigma_z^2}{2\pi} \left(\frac{489}{400} + \cos w + \frac{1}{2} \cos 2w \right) \\ &= \frac{\sigma_z^2}{2\pi} \left(\frac{489}{400} + \cos w + \cos^2 w - \frac{1}{2} \right) \\ &= \frac{\sigma_z^2}{2\pi} \left(\cos^2 w + \cos w + \frac{289}{400} \right) \quad (w \in (-\pi, \pi]). \end{aligned}$$

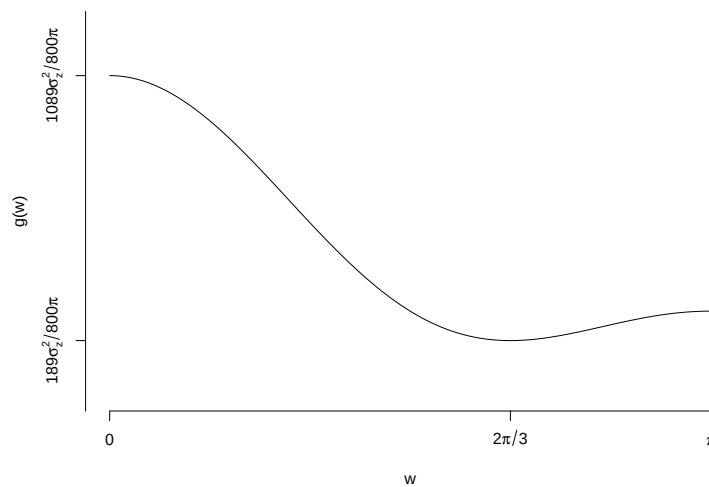
- (c) Sketch the spectral density function from (b) for values of the frequency, w , where $w \in [0, \pi]$.

SOLUTION

Re-write the spectral density function as

$$g(w) = \frac{\sigma_z^2}{2\pi} \left(\left(\cos w + \frac{1}{2} \right)^2 + \frac{189}{400} \right).$$

The plot is shown below.



- (d) The time series $\{Y_t : t \in \mathbb{Z}\}$ is a linear filtered version of $\{X_t : t \in \mathbb{Z}\}$ where

$$Y_t = \frac{1}{2} (X_t + X_{t-1})$$

Determine the spectral density function of $\{Y_t : t \in \mathbb{Z}\}$.

SOLUTION

The transfer function is

$$h(w) = \sum_{j=-\infty}^{\infty} b_j e^{-i w j}$$

where

$$b_j = \begin{cases} \frac{1}{2} & j = 0; \\ \frac{1}{2} & j = 1; \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$h(w) = \frac{1}{2} (1 + e^{-i w}).$$

The squared gain is

$$\begin{aligned}
 |h(w)|^2 &= \left| \frac{1}{2} (1 + e^{-iw}) \right|^2 \\
 &= \frac{1}{2} (1 + e^{-iw}) \times \frac{1}{2} (1 + e^{iw}) \\
 &= \frac{1}{4} (2 + 2 \cos w) \\
 &= \frac{1}{2} (1 + \cos w).
 \end{aligned}$$

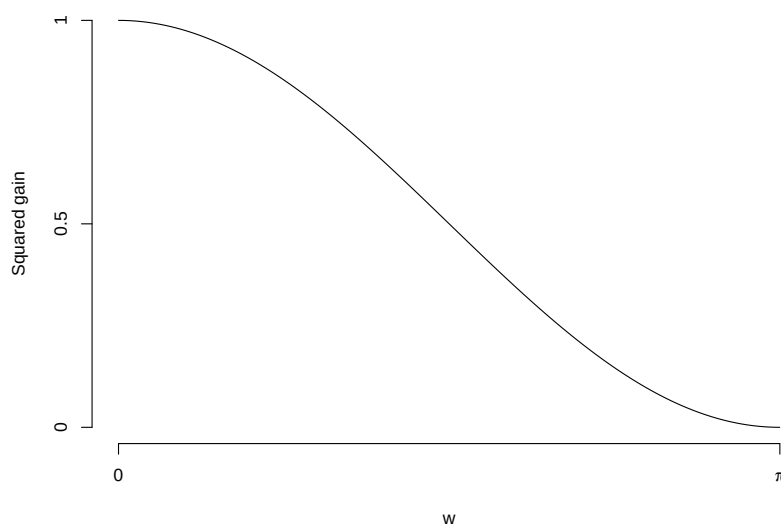
The spectral density function of $\{Y_t : t \in \mathbb{Z}\}$ is

$$\begin{aligned}
 g_Y(w) &= |h(w)|^2 g_X(w) \\
 &= \frac{1}{2} (1 + \cos w) \left[\frac{\sigma_z^2}{2\pi} \left(\left(\cos w + \frac{1}{2} \right)^2 + \frac{189}{400} \right) \right] \\
 &= \frac{\sigma_z^2}{4\pi} (1 + \cos w) \left(\left(\cos w + \frac{1}{2} \right)^2 + \frac{189}{400} \right) \quad (w \in (-\pi, \pi]).
 \end{aligned}$$

- (e) Sketch the squared gain for the linear filter described in (d). Hence, describe briefly the effect of this linear filter on the original process $\{X_t : t \in \mathbb{Z}\}$.

SOLUTION

A sketch of the squared gain is shown below. Contributions from all frequencies



are reduced, with those from higher frequencies (close to π) reduced more than lower frequencies. Consequently, the transformed process will be smoother.

3. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 . You should show your working and/or justification clearly in all answers that involve calculations or derivations.

The (weakly) stationary process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = 200 - 0.85X_{t-1} + 0.6X_{t-2} + Z_t - 0.25Z_{t-1}.$$

- (a) Determine the mean of $\{X_t : t \in \mathbb{Z}\}$

SOLUTION

Taking the mean of both sides we see

$$\begin{aligned} \mathbb{E}(X_t) &= 200 - 0.85\mathbb{E}(X_{t-1}) + 0.6\mathbb{E}(X_{t-2}) + \mathbb{E}(Z_t) - 0.25\mathbb{E}(Z_{t-1}) \\ \implies \mathbb{E}(X_t) &= 200 - 0.85\mathbb{E}(X_{t-1}) + 0.6\mathbb{E}(X_{t-2}) \end{aligned}$$

Because the process is (weakly) stationary we have $\mathbb{E}(X_t) = \mathbb{E}(X_{t-1}) = \mathbb{E}(X_{t-2})$ and, therefore

$$\begin{aligned} 1.25\mathbb{E}(X_t) &= 200 \\ \implies \mathbb{E}(X_t) &= 160. \end{aligned}$$

- (b) Hence show that $\{X_t : t \in \mathbb{Z}\}$ can be written in the form

$$X_t - \mu = a(X_{t-1} - \mu) + b(X_{t-2} - \mu) + Z_t - 0.25Z_{t-1}$$

where a , b and μ are real-valued constants.

SOLUTION

We see that

$$\begin{aligned} X_t - 160 &= (200 - 0.85X_{t-1} + 0.6X_{t-2} + Z_t - 0.25Z_{t-1}) - 160 \\ \implies X_t - 160 &= -0.85X_{t-1} + 0.6X_{t-2} + 40 + Z_t - 0.25Z_{t-1} \\ \implies X_t - 160 &= -0.85(X_{t-1} - 160) + 0.6(X_{t-2} - 160) - 0.25Z_{t-1} + Z_t \end{aligned}$$

as required.

(2 MARKS)

- (c) The process $\{X_t : t \in \mathbb{Z}\}$ (considered in (a) and (b)) is assumed to be the annual number of passengers who use a particular railway station (in 1,000s). The following table shows annual numbers of passengers for the years 2022–2025:

Year	2022	2023	2024	2025
No. of passengers (thousands)	159	162	156	168

Using this information, produce an optimal forecast for the number of passengers who use the station for the year 2026. Here, the term ‘optimal’ implies that the

forecasted value is that which minimises the mean square prediction error at the corresponding time point.

SOLUTION

We consider $Y_t = X_t - 160$ and, hence

$$Y_t = -0.85Y_{t-1} + 0.6Y_{t-2} - 0.25Z_{t-1} + Z_t.$$

Supposing that the series is observed to time n . The k -step ahead predictor is

$$\hat{Y}_{n+k} = -0.85Y_{n+k-1} + 0.6Y_{n+k-2} - 0.25[Z_{n+k-1}] + [Z_{n+k}]$$

where $[Z_t]$ is defined

$$[Z_t] = \begin{cases} 0 & t > n; \\ Z_t & t \leq n. \end{cases}$$

Here

$$\begin{aligned} \hat{Y}_{2026} &= -0.85\hat{Y}_{2025} + 0.6\hat{Y}_{2024} - 0.25Z_{2025} \\ \implies \hat{Y}_{2026} &= -0.85Y_{2025} + 0.6Y_{2024} - 0.25Z_{2025} \end{aligned}$$

To estimate Z_{2025} we consider

$$\begin{aligned} Y_t + 0.85Y_{t-1} - 0.6Y_{t-2} &= -0.25Z_{t-1} + Z_t \\ \implies (1 + 0.85B - 0.6B^2) Y_t &= (1 - 0.25B) Z_t \end{aligned}$$

and, therefore

$$Z_t = \frac{(1 + 0.85B - 0.6B^2)}{1 - 0.25B} Y_t.$$

Let $f(B) = (1 - 0.25B)^{-1}$ then

$$\begin{aligned} f'(B) &= 0.25(1 - 0.25B)^{-2} \\ f''(B) &= 2(0.25^2)(1 - 0.25B)^{-3} \end{aligned}$$

Generally, if $f^{(k)}(B)$ is the k^{th} derivative of $f(B)$ then

$$f^{(k)}(B) = k!(0.25^k)(1 - 0.25B)^{-(k+1)}.$$

The Taylor series expansion of $f(B)$ about 0 is

$$\begin{aligned} f(B) &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} B^k \\ &= \sum_{k=0}^{\infty} (0.25^k) B^k. \end{aligned}$$

Then

$$\begin{aligned} Z_t &= (1 + 0.85B - 0.6B^2) \sum_{k=0}^{\infty} (0.25^k) B^k Y_t \\ &= \sum_{k=0}^{\infty} (0.25^k) Y_{t-k} + 0.85 \sum_{k=1}^{\infty} (0.25^{k-1}) Y_{t-k} - 0.6 \sum_{k=2}^{\infty} (0.25^{k-2}) Y_{t-k} \end{aligned}$$

Hence, the estimate of Z_{2025} is

$$\begin{aligned} \hat{Z}_{2025} &= Y_{2025} + (0.25 + 0.85) Y_{2024} + (0.25^2 + 0.85(0.25) - 0.6) Y_{2023} \\ &\quad + (0.25^3 + 0.85(0.25^2) - 0.6(0.25)) Y_{2022}. \\ &= Y_{2025} + 1.1 Y_{2024} - 0.325 Y_{2023} - 0.08125 Y_{2022}. \end{aligned}$$

The observed values of $\{Y_t\}$ are

Year	2022	2023	2024	2025
No. of passengers (hundreds of thousands)	-1	2	-4	8

Then

$$\begin{aligned} \hat{Z}_{2025} &= 8 + 1.1(-4) - 0.325(2) - 0.08125(-1) \\ &= 3.03125. \end{aligned}$$

Hence

$$\begin{aligned} \hat{Y}_{2026} &= -0.85 Y_{2025} + 0.6 Y_{2024} - 0.25 Z_{2025} \\ &= -0.85(8) + 0.6(-4) - 0.25(3.03125) \\ &= -9.957812. \end{aligned}$$

Finally

$$\hat{X}_{2026} = 160 - 9.957812 = 150.0422.$$

4. Throughout this question the process $\{Z_t : t \in \mathbb{Z}\}$ is a white noise process with mean zero and variance σ_z^2 . You should show your working and/or justification clearly in all answers that involve calculations or derivations.

(a) The process $\{X_t : t \in \mathbb{Z}\}$ is defined

$$X_t = (a + bt)s_t + ct + Z_t$$

where a , b and c are non-zero, real-valued constants and s_t is a seasonal term of period 4, such that $s_t = s_{t-4}$ for all t .

(i) Show that $\{X_t : t \in \mathbb{Z}\}$ is not a stationary process.

(SIMILAR TO SEEN MATERIAL)

SOLUTION

Consider $\mathbb{E}(X_t)$ where

$$\begin{aligned}\mathbb{E}(X_t) &= \mathbb{E}[(a + bt)s_t + ct + Z_t] \\ &= (a + bt)s_t + ct.\end{aligned}$$

Clearly $\mathbb{E}(X_t)$ depends on t and therefore $\{X_t : t \in \mathbb{Z}\}$ is not stationary.

(ii) Suppose that ∇_k is a difference operator that can be applied a process, where

$$\nabla_k = 1 - B^k$$

for $k \in \mathbb{N}$ and B denotes the backward shift operator.

Using difference operators of this form, show how the process $\{X_t : t \in \mathbb{Z}\}$ can be transformed to be a (weakly) stationary process.

SOLUTION

Because of the seasonality of period 4, a sensible start would be to take a difference at lag 4.

$$\begin{aligned}\nabla_4 X_t &= X_t - X_{t-4} \\ &= [(a + bt)s_t + ct + Z_t] - [(a + b(t-4))s_{t-4} + c(t-4) + Z_{t-4}] \\ &= (a + bt)(s_t - s_{t-4}) + 4bs_{t-4} + 4c + Z_t - Z_{t-4} \\ &= 4(bs_{t-4} + c) + Z_t - Z_{t-4}.\end{aligned}$$

Then

$$\begin{aligned}\nabla_4^2 X_t &= [4(bs_{t-4} + c) + Z_t - Z_{t-4}] - [4(bs_{t-8} + c) + Z_{t-4} - Z_{t-8}] \\ &= Z_t - 2Z_{t-4} + Z_{t-8}.\end{aligned}$$

Clearly, $\nabla_4^2 X_t$ is a (weakly) stationary process.

- (b) The process $\{U_t : t \in \mathbb{Z}\}$ is a (weakly) stationary process with mean μ , variance σ_u^2 and autocorrelation function $\rho_U(h)$. The process $\{Y_t : t \in \mathbb{Z}\}$ is defined

$$Y_t = \frac{1}{2}U_t + \frac{1}{3}Z_{t-1} + Z_t$$

and the processes $\{U_t : t \in \mathbb{Z}\}$ and $\{Z_t : t \in \mathbb{Z}\}$ are independent.

- (i) Determine the autocorrelation function of $\{Y_t : t \in \mathbb{Z}\}$, in terms of $\rho_U(h)$ and the process parameters.

SOLUTION

The autocovariance function for $\{Y_t\}$ at lag h is

$$\begin{aligned} \gamma_Y(h) &= \text{Cov}(Y_t, Y_{t+h}) \\ &= \text{Cov}\left(\frac{1}{2}U_t + \frac{1}{3}Z_{t-1} + Z_t, \frac{1}{2}U_{t+h} + \frac{1}{3}Z_{t+h-1} + Z_{t+h}\right) \\ &= \left(\frac{1}{2}\right)^2 \text{Cov}(U_t, U_{t+h}) + \left(\frac{1}{3}\right)^2 \text{Cov}(Z_{t-1}, Z_{t+h-1}) + \frac{1}{3} \text{Cov}(Z_{t-1}, Z_{t+h}) \\ &\quad + \frac{1}{3} \text{Cov}(Z_t, Z_{t+h-1}) + \text{Cov}(Z_t, Z_{t+h}) \\ &= \frac{1}{4} \text{Cov}(U_t, U_{t+h}) + \frac{1}{9} \text{Cov}(Z_{t-1}, Z_{t+h-1}) + \frac{1}{3} \text{Cov}(Z_{t-1}, Z_{t+h}) \\ &\quad + \frac{1}{3} \text{Cov}(Z_t, Z_{t+h-1}) + \text{Cov}(Z_t, Z_{t+h}) \end{aligned}$$

If $h = 0$

$$\begin{aligned} \gamma_Y(h) &= \text{Var}(Y_t) \\ &= \frac{1}{4} \text{Var}(U_t) + \frac{1}{9} \text{Var}(Z_{t-1}) + \text{Var}(Z_t) \\ &= \frac{1}{4} \sigma_u^2 + \frac{10}{9} \sigma_z^2. \end{aligned}$$

If $h = \pm 1$

$$\begin{aligned} \gamma_Y(h) &= \frac{1}{4} \gamma_U(1) + \frac{1}{3} \sigma_z^2 \\ &= \frac{1}{4} \sigma_u^2 \rho_U(1) + \frac{1}{3} \sigma_z^2. \end{aligned}$$

If $|h| > 1$

$$\begin{aligned} \gamma_Y(h) &= \frac{1}{4} \gamma_U(h) \\ &= \frac{1}{4} \sigma_u^2 \rho_U(h). \end{aligned}$$

Then, the autocorrelation function is

$$\rho_Y(h) = \begin{cases} 1 & h = 0; \\ \frac{\frac{1}{4}\sigma_u^2\rho_U(1)+\frac{1}{3}\sigma_z^2}{\frac{1}{4}\sigma_u^2+\frac{10}{9}\sigma_z^2} & |h| = 1; \\ \frac{\frac{1}{4}\sigma_u^2\rho_U(h)}{\frac{1}{4}\sigma_u^2+\frac{10}{9}\sigma_z^2} & |h| \geq 2. \end{cases}$$

(ii) Is $\{Y_t : t \in \mathbb{Z}\}$ a (weakly) stationary process? Justify your answer.

SOLUTION

We see that

$$\begin{aligned} \mathbb{E}(Y_t) &= \frac{1}{2}\mathbb{E}(U_t) + \frac{1}{3}\mathbb{E}(Z_{t-1}) + \mathbb{E}(Z_t) \\ &= \frac{\mu}{2}. \end{aligned}$$

so the mean of $\{Y_t\}$ is constant over time.

In addition

$$\text{Var}(Y_t) = \frac{1}{4}\sigma_u^2 + \frac{10}{9}\sigma_z^2$$

which is also constant over time.

Finally, from (i), $\rho_U(h)$ does not depend on t because $\{U_t\}$ is a (weakly) stationary process and therefore $\rho_Y(h)$ does not depend on t .

Hence $\{Y_t : t \in \mathbb{Z}\}$ is a (weakly) stationary process.

(iii) By considering convergence in mean square, show that $\{Y_t : t \in \mathbb{Z}\}$ converges to the MA(1) process

$$Y_t = \frac{\mu}{2} + \frac{1}{3}Z_{t-1} + Z_t$$

as $\sigma_u^2 \rightarrow 0$.

SOLUTION

Let V_t be the process

$$V_t = \frac{\mu}{2} + \frac{1}{3}Z_{t-1} + Z_t.$$

For convergence in mean square, we must show that

$$\lim_{\sigma_u^2 \rightarrow 0} \mathbb{E}[(Y_t - V_t)^2] = 0.$$

Consider

$$\mathbb{E}[(Y_t - V_t)^2] = \mathbb{E}\left[\left(\frac{1}{2}U_t + \frac{1}{3}Z_{t-1} + Z_t - \frac{\mu}{2} - \frac{1}{3}Z_{t-1} - Z_t\right)^2\right]$$

$$\begin{aligned} &= \mathbb{E} \left[\left(\frac{1}{2} U_t - \frac{\mu}{2} \right)^2 \right] \\ &= \frac{1}{4} \mathbb{E} [(U_t - \mu)^2] \\ &= \frac{\sigma_u^2}{4}. \end{aligned}$$

Clearly

$$\lim_{\sigma_u^2 \rightarrow 0} \frac{\sigma_u^2}{4} = 0$$

and it follows that Y_t converges to the process

$$Y_t = \frac{\mu}{2} + \frac{1}{3} Z_{t-1} + Z_t.$$